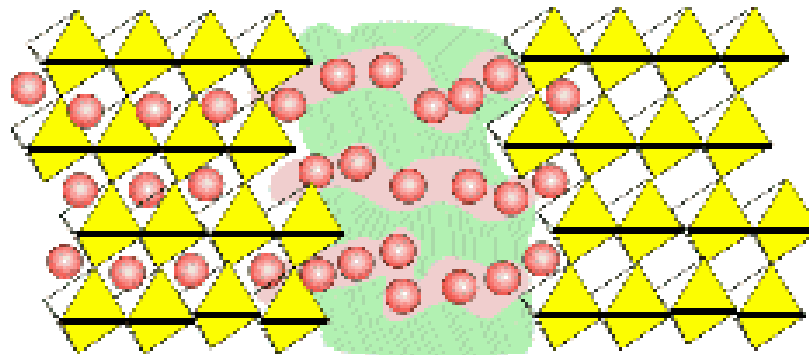


Electrochemistry in Solids

Thermodynamics and Kinetics

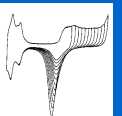


*Electrochemistry at
the other side of
the interface*



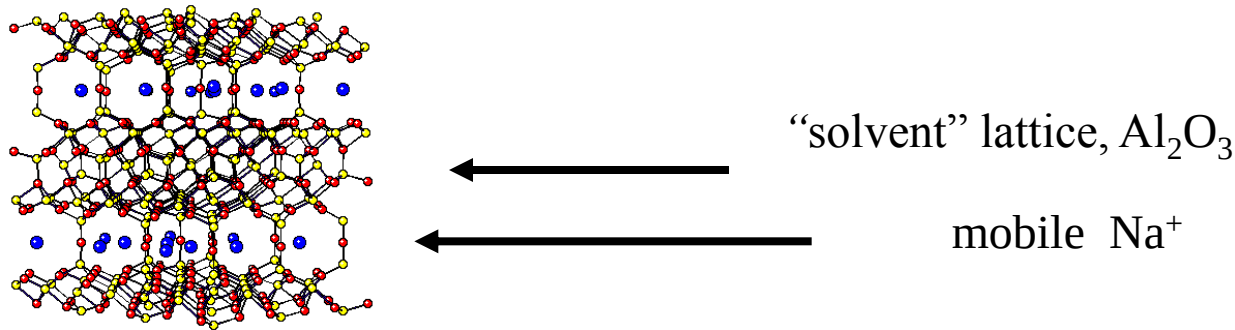
SEPTEMBER 17-20, 2018

SALA D'ACTES CARLES MIRAVITLLES
ICMAB-CSIC, Bellaterra, Spain

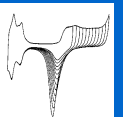
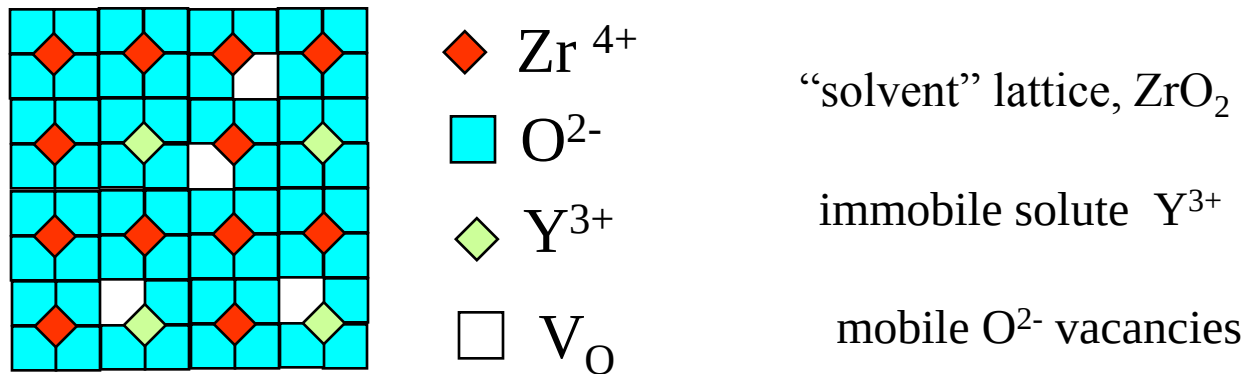


Ions can be mobile in a solid lattice - as in Solid Electrolytes

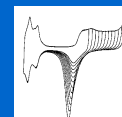
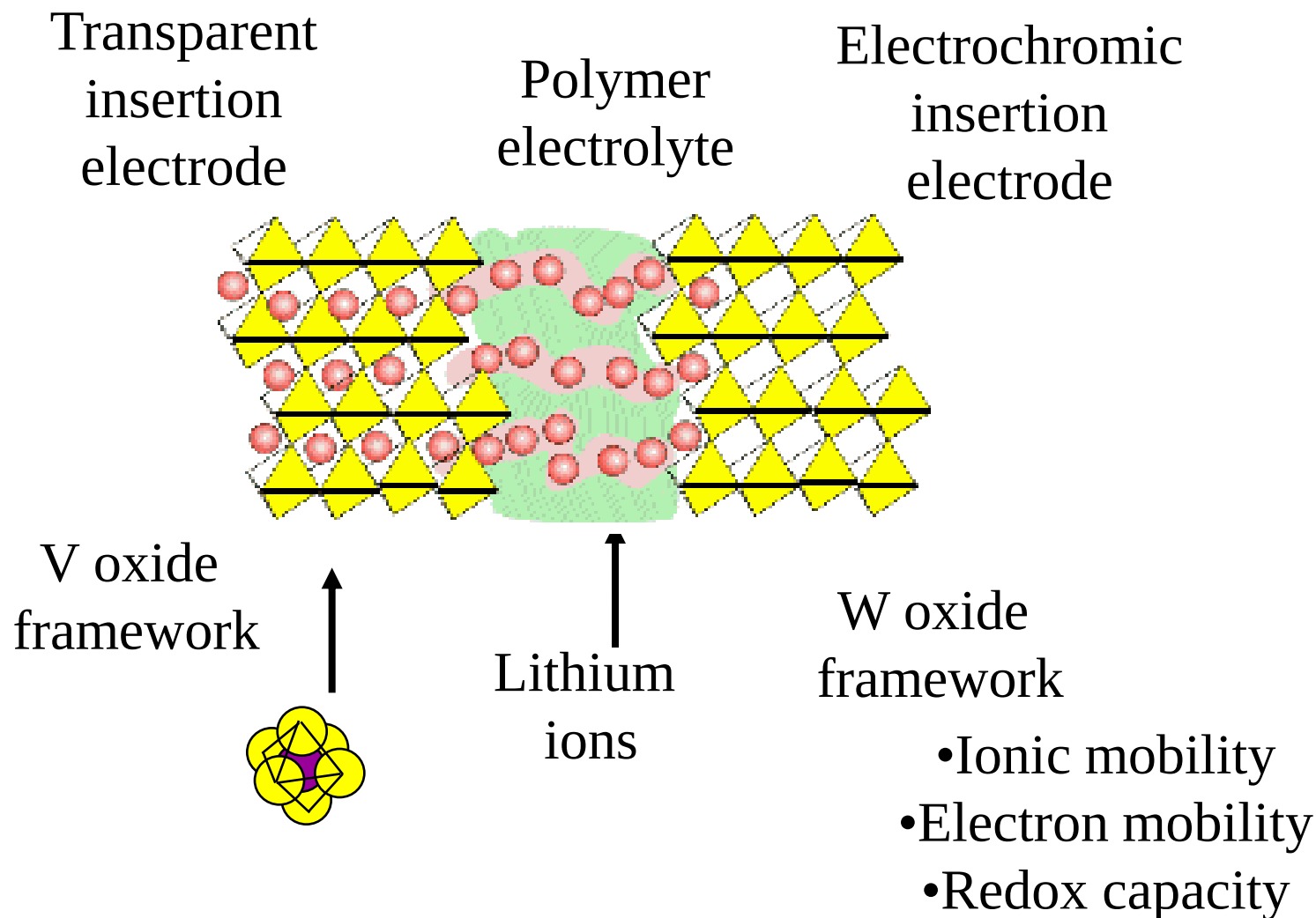
e.g. Interstitial type, β -Alumina



e.g. Vacancy type, Yttria stabilized zirconia (YSZ)

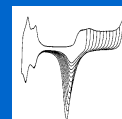
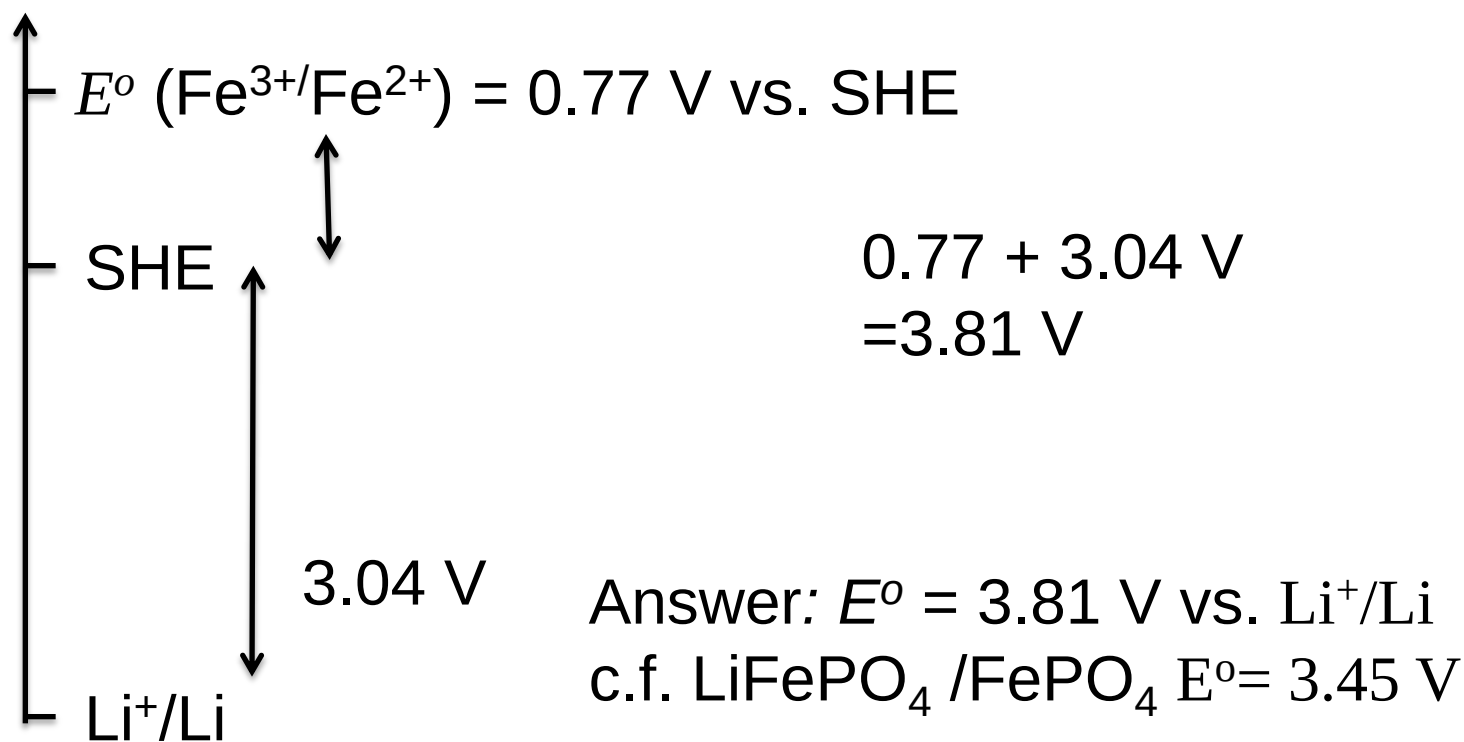


Insertion Electrodes in Cells

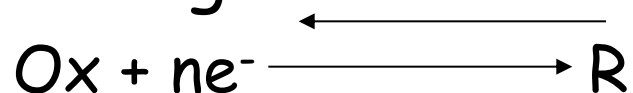


Conversion of potentials

For a potential value equal to 0.77 V vs SHE, convert it to the Li scale taking into account that Li equals -3.04 V vs. SHE



THERMODYNAMICS The Classical Nernst equation for single reactant and product



$$E_e = E^0 - \frac{RT}{nF} \ln \frac{a_R}{a_{\text{Ox}}} = E^0 + \frac{RT}{nF} \ln \frac{a_{\text{Ox}}}{a_R}$$

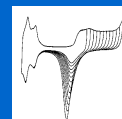
E_e : equilibrium potential

E^0 : standard potential

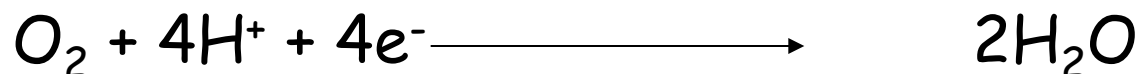
a_{Ox}, a_R : activities of oxidized and reduced species

For *ideal* solutions $a = c/c^0$, e.g. where $c^0 = 1\text{M}$

The standard potential, E^0 , equals the equilibrium potential under conditions of unity activity of all reactants and products



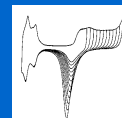
The General Nernst equation



$$E_e = E^0 + \frac{RT}{nF} \ln \frac{\prod a_{Ox}}{\prod a_R} = E^0 + \frac{RT}{4F} \ln \frac{a_{O_2} a_{H^+}^4}{a_{H_2O}^2}$$

E_e : equilibrium potential

E^0 : standard potential

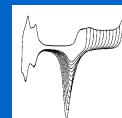


What is the activity?

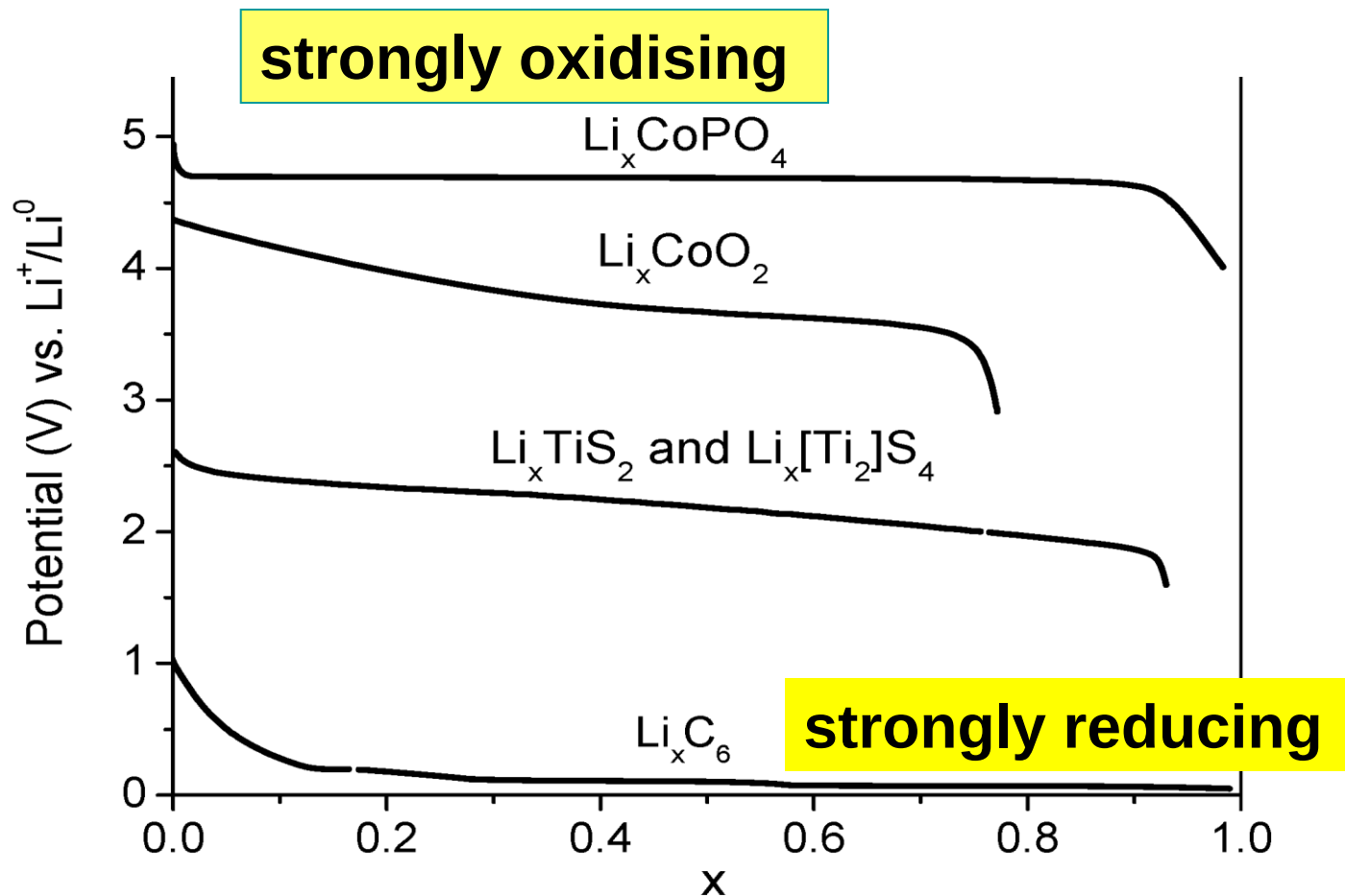
- For *ideal* solutions $a = c/c^0$, where c^0 could be 1M
- *activity* expresses the *effective* concentration, pressure, or any other deviation from the standard state.
- For real solutions the activity is obtained from the concentration through the activity coefficient γ which describes a surprised or enhanced reactivity for values less than or greater than 1.

$$a = \gamma (c/c^0)$$

- *If the concentration of a component is constant, e.g. due to a 2-phase equilibrium, we can set the standard state ΔG° so that $a = 1$.*
- In solid state electrochemistry we often use the thermodynamic factor ($d \ln a / d \ln c$) to express non ideality c.f. ideal solutions in which $(d \ln a / d \ln c) = 1$ (see later)



The potential-composition relation

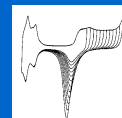


composition range

0

[red]

~50 M !!!



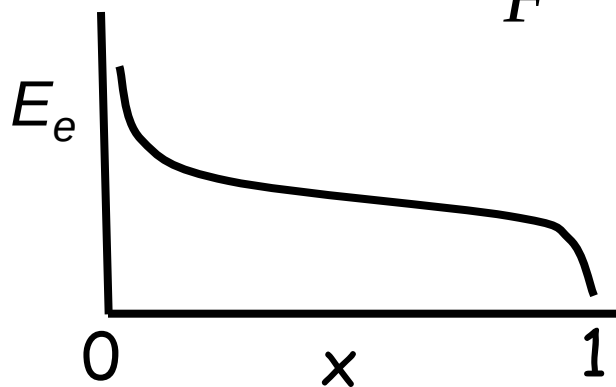
The potential-composition relation

For the reaction:



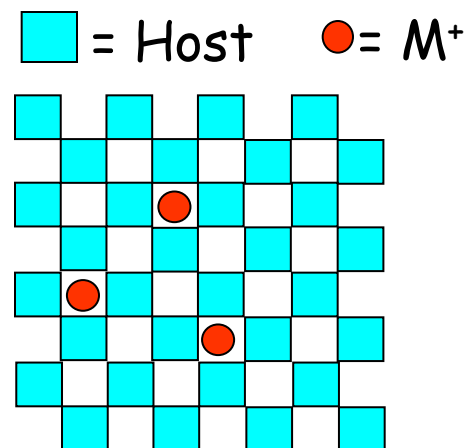
The lattice gas model results in a Nernst-like expression:

$$E_e = E^0 - \frac{RT}{F} \ln \left(\frac{x}{1-x} \right)$$

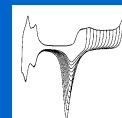


Host

$\text{Li}_x(\text{Host})$



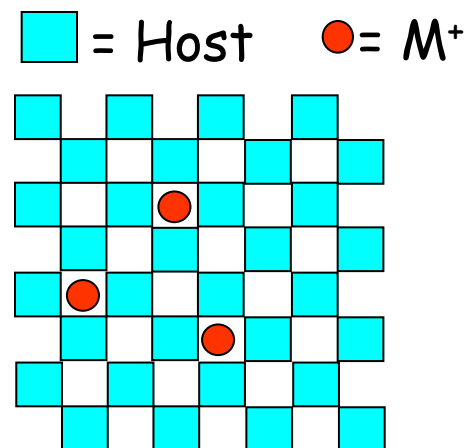
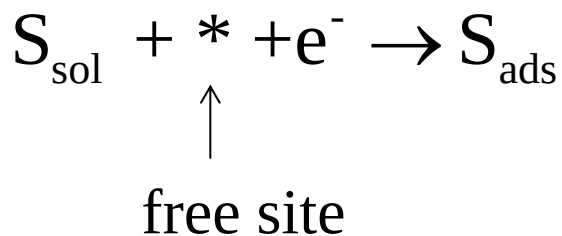
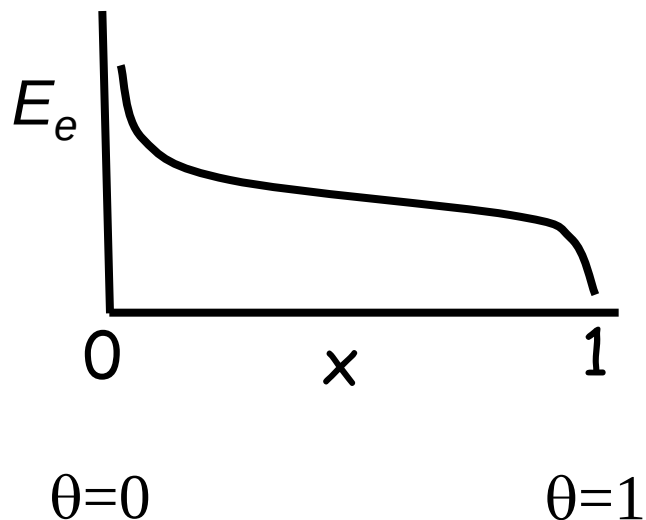
(The Lattice Gas Model)



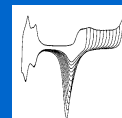
The potential-composition relation

Also the Langmuir Isotherm has

$$E_e = E^0 - \frac{RT}{F} \ln \left(\frac{x}{1-x} \right)$$



(The Lattice Gas Model)

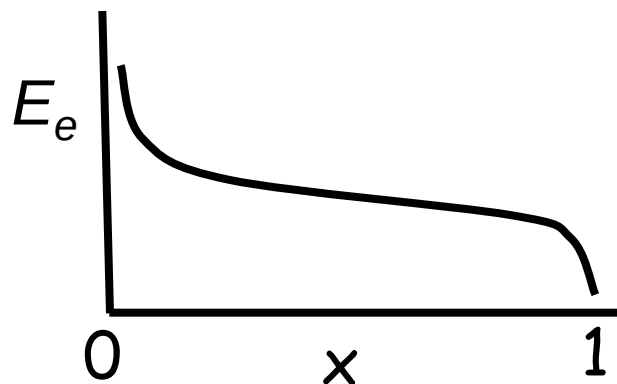


The potential-composition relation

Here comes non-ideality:

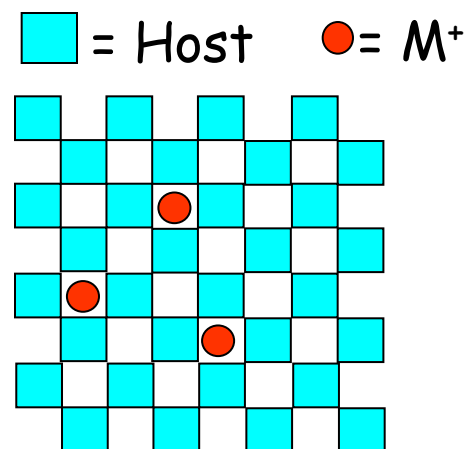
$$E_e = E^0 - \frac{RT}{F} \ln \left(\frac{x}{1-x} \right) + kx$$

↑
interactions

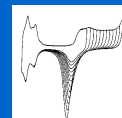


Host

$\text{Li}_x(\text{Host})$



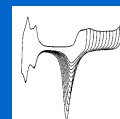
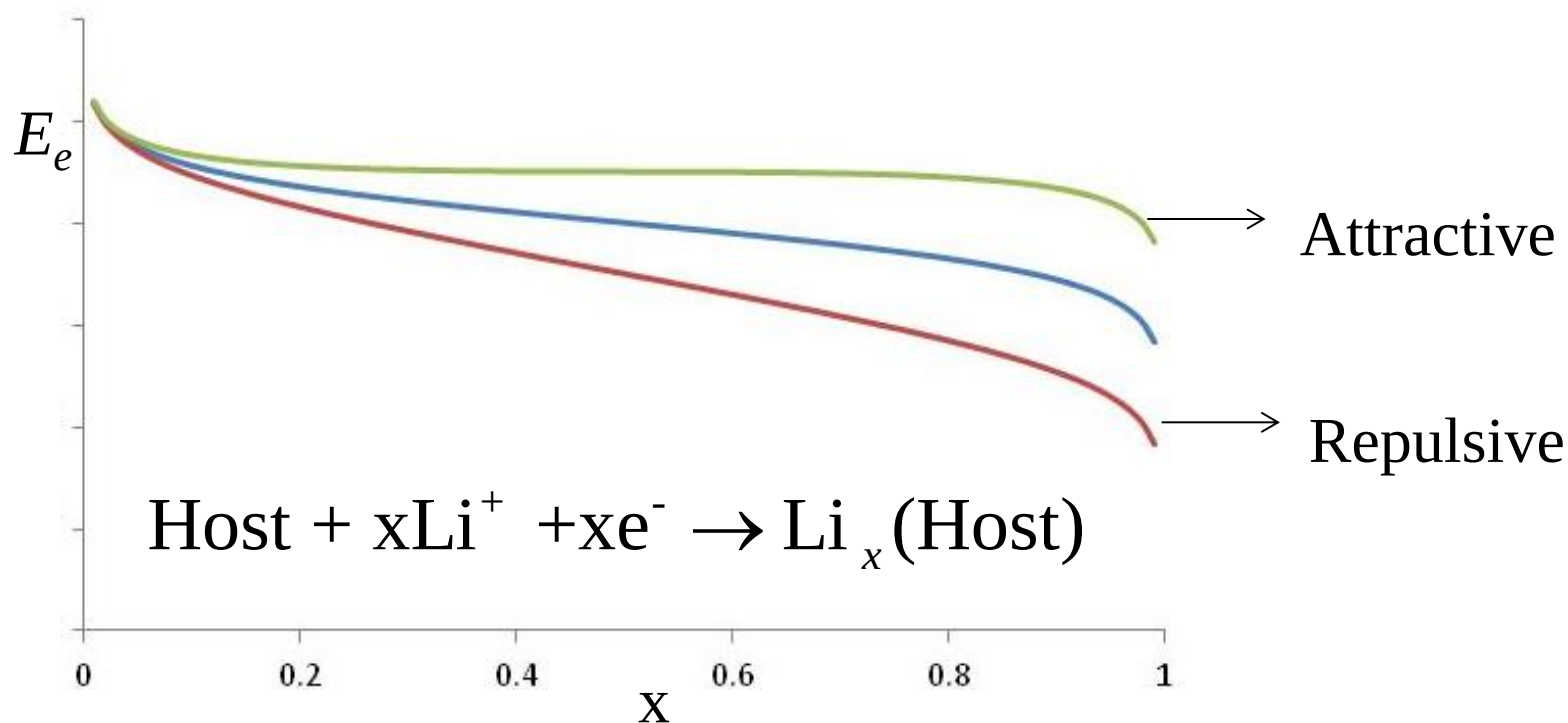
(The Lattice Gas Model)



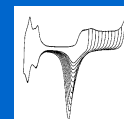
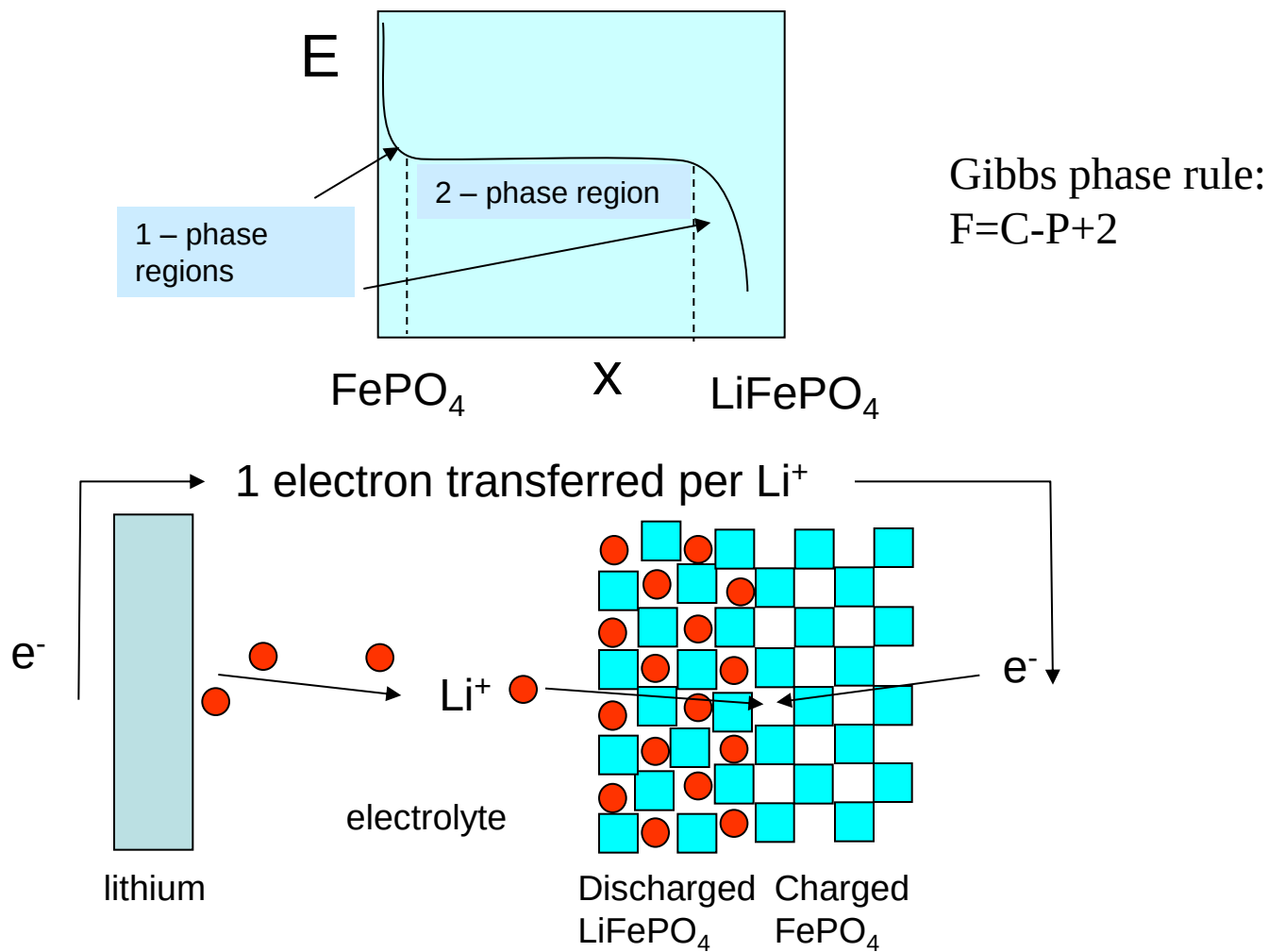
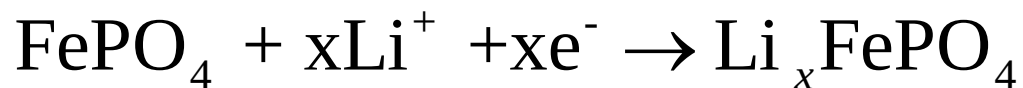
The potential-composition relation

$$E_e = E^0 - \frac{RT}{F} \ln \left(\frac{x}{1-x} \right) + kx$$

↖ interactions

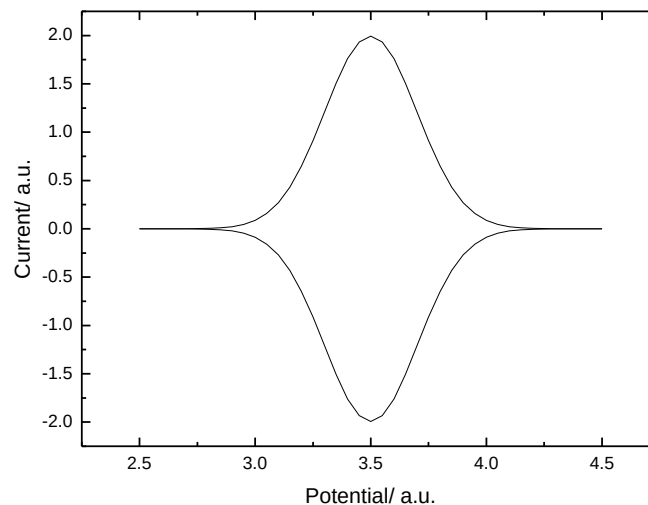
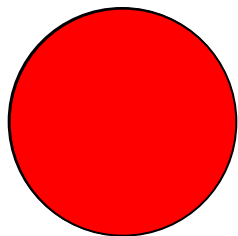


Attractive interactions can lead to phase separation

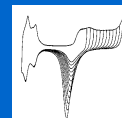
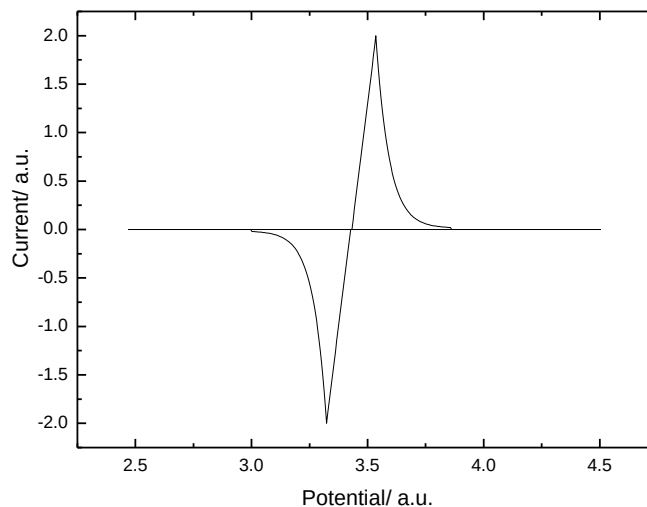
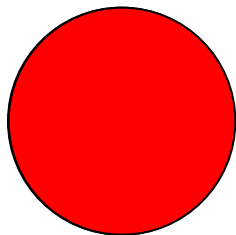


Slow cyclic voltammetry of **battery electrodes**

1 Phase



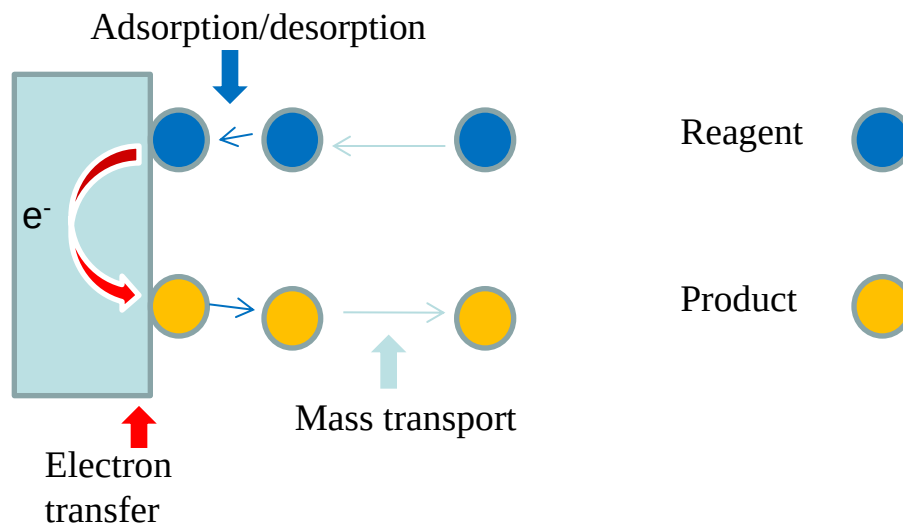
2 Phase



Solid redox reactions need solid diffusion

SOLUTION

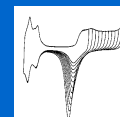
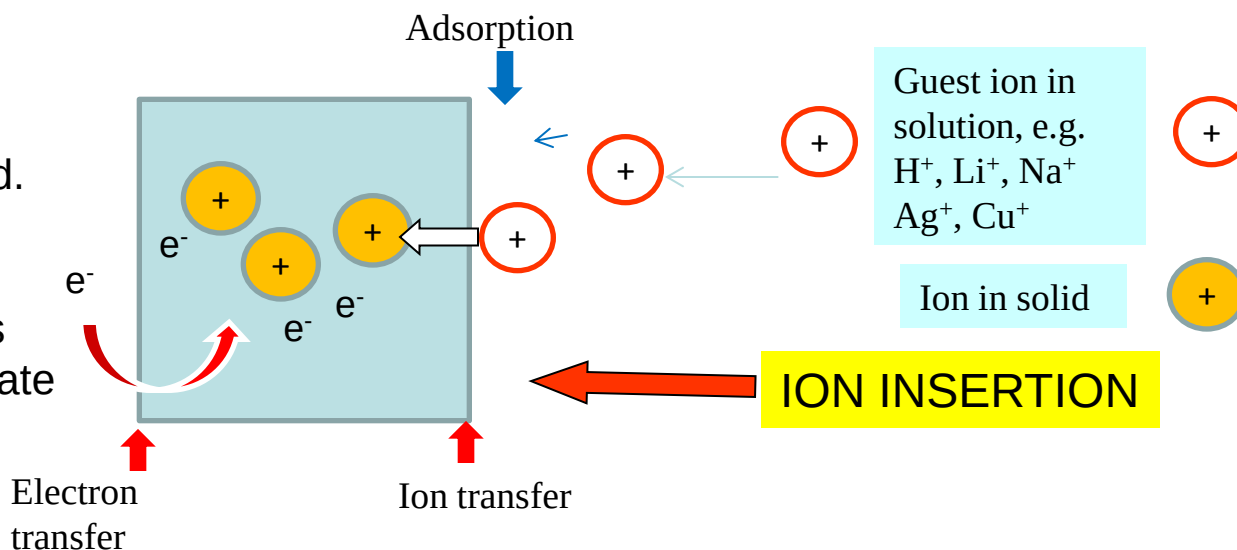
The redox reagent is initially in the solution. It diffuses to the interface, reacts, then returns to the solution,



SOLID

The reagent is a solid.

The guest ion enters the solid to compensate the charge.



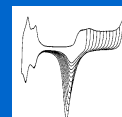
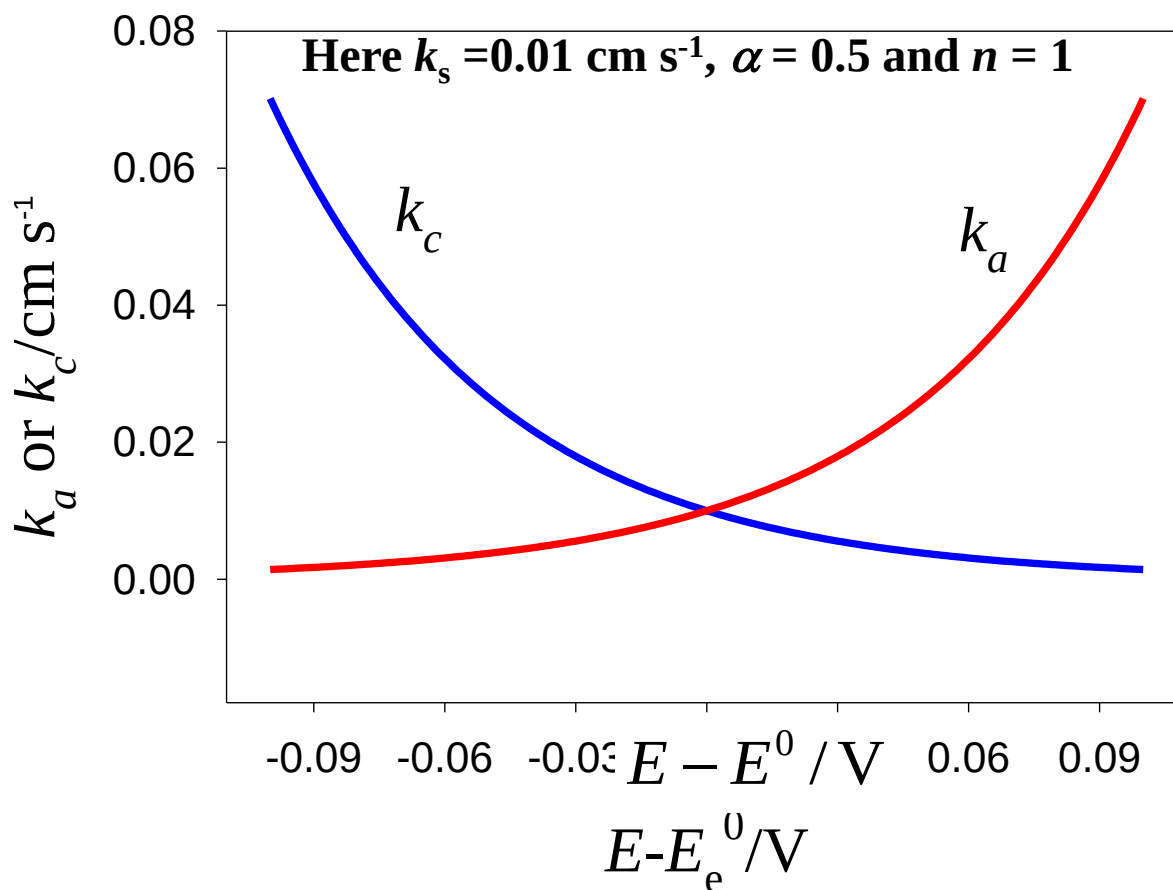
Kinetics in Solid State Electrochemistry

Potential Dependent Rate Constant

$$k_c = k_s e^{\frac{-\alpha_c nF(E-E^0)}{RT}}$$

$$\text{Rate} = kc$$

$$k_a = k_s e^{\frac{\alpha_a nF(E-E^0)}{RT}}$$



Butler Volmer Kinetics

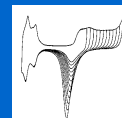
The kinetics of electron transfer shown above are expressed by the **Butler Volmer equation**

$$j = j_0 \left[\exp \left(\frac{\alpha_a n F}{RT} \eta \right) - \exp \left(\frac{-\alpha_c n F}{RT} \eta \right) \right]$$

where

Overpotential $\eta = E - E_e$

j_0 = exchange current at equilibrium

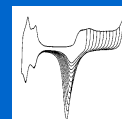


Relevance of Butler -Volmer Kinetics for batteries

$$j = j_0 \left[\exp \left(\frac{\alpha_a n F}{RT} \eta \right) - \exp \left(\frac{-\alpha_c n F}{RT} \eta \right) \right]$$

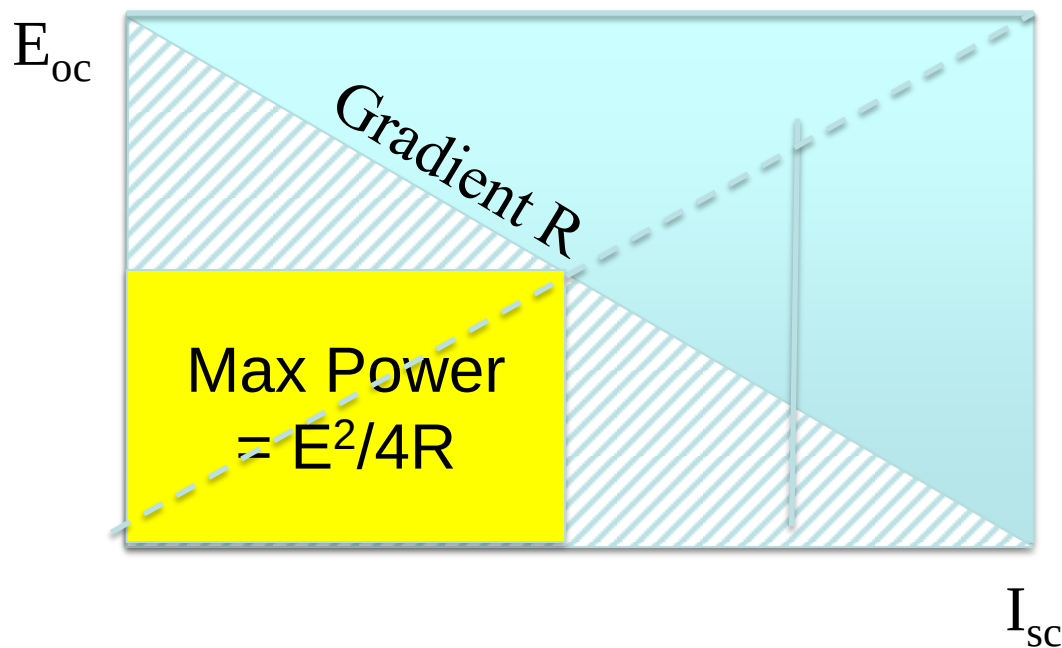
For a simple one-electron transfer reaction $\alpha_a + \alpha_c = 1$

- Typically the current should increase 10x for each 30 mV increase in overpotential - i.e. kinetics can be accelerated at low cost!
- j_0 is proportional to c , so the rate slows toward zero as reactant depletes at end of discharge - an essential feature of a realistic model.



Also - Resistance limitations

- R = ionic resistance of the electrolyte + electronic resistance of the electrode



- We need better electrolytes!

